3D Gaussian Splatting: Implementation, Anisotropic Correlation Analysis Project Report

Diya Kasera AU2220135
STA 306 Computational Statistics
Based on Kerbl et al. (2023)

Table of Contents

1. Introduction

2. Methodology

3. Anisotropic correlation analysis

4. Proposed improvements for anisotropic capture

5. Experimental setup

6. Results

7. Limitations

8. Conclusion

9. References

---

1. Introduction

Radiance fields have transformed the task of rendering photorealistic images of a scene from novel viewpoints (novel-view synthesis). Neural Radiance Fields (NeRF) achieve state-of-the-art visual quality, but at the cost of speed: hours of training and sub-real-time rendering.

3D Gaussian Splatting (3DGS) by Kerbl et al. (2023) addresses these issues by explicitly modelling the scene as a differentiable splatting of millions of anisotropic 3D Gaussians. Through alpha-blending of Gaussian primitives on the image plane, 3DGS eliminates the need to evaluate a neural network along a ray, allowing real-time rendering at 1080p (≥30 fps) with similar quality.

This report offers a full implementation of 3DGS on a custom dataset of 141 photographs, assesses the trained model's capacity to represent anisotropic spatial correlation, and proposes enhancements to better capture anisotropy.

---

2. Methodology

2.1 Scene Representation

Each scene is represented as a set of N learnable 3D Gaussians. The influence of the i-th Gaussian at a point $\mathbf{x} \in \mathbb{R}^3$ is defined by:

$$G_i(\mathbf{x}) = \exp\left(-\frac{1}{2}(\mathbf{x} - \boldsymbol{\mu}_i)^\top \boldsymbol{\Sigma}_i^{-1}(\mathbf{x} - \boldsymbol{\mu}_i)\right)$$

where $\boldsymbol{\mu}_i \in \mathbb{R}^3$ is the Gaussian centre (mean position) and $\boldsymbol{\Sigma}_i \in \mathbb{R}^{3\times3}$ is the covariance matrix, encoding the shape, scale, and orientation of the primitive. Each Gaussian additionally carries opacity $\alpha_i \in [0, 1]$ and a view-dependent colour $\mathbf{c}_i$ represented via Spherical Harmonic (SH) coefficients.

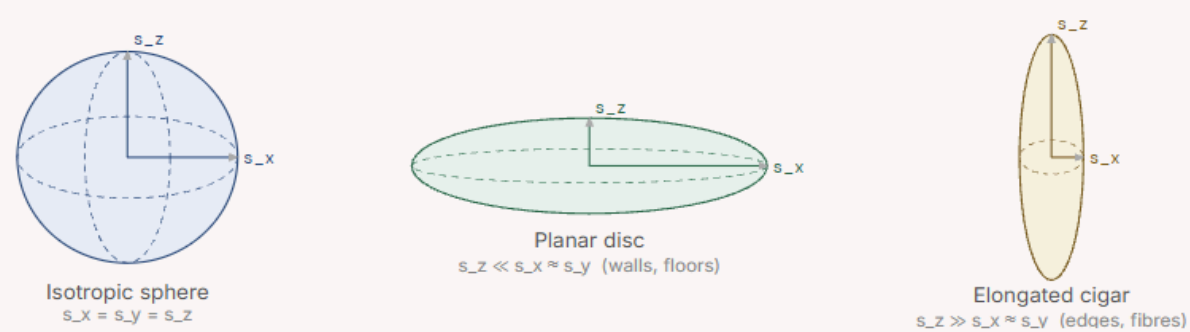

Figure 1. *The three primary anisotropic Gaussian shapes arising from independent scale parameters (s_x, s_y, s_z). Real scenes simultaneously employ all three modes to represent surfaces, edges, and volumetric regions.*

2.2 Covariance Parameterisation and Anisotropy

A key design choice in Kerbl et al. (2023) is that the covariance matrix Σ must remain symmetric positive semi-definite (PSD) throughout optimisation to produce valid Gaussians. Rather than directly optimising Σ (which gradient descent could render non-PSD), the matrix is decomposed as:

$$\mathbf{\Sigma} = \mathbf{R}\,\mathbf{S}\,\mathbf{S}^\top\mathbf{R}^\top$$

where $\mathbf{S} = \mathrm{diag}(s_x,\ s_y,\ s_z)$ is a diagonal scaling matrix with three independent positive scale parameters, and $\mathbf{R}$ is a rotation matrix parameterised by a unit quaternion $\mathbf{q} = (q_w, q_x, q_y, q_z) \in \mathbb{R}^4,\ \|\mathbf{q}\| = 1$. This factorisation guarantees $\mathbf{\Sigma} \succeq 0$ by construction. The three eigenvalues of $\mathbf{\Sigma}$ — the squares of the diagonal entries of $\mathbf{S}$ — encode variance along each principal axis. Unequal eigenvalues produce anisotropic ellipsoids.

This factorisation is the source of anisotropy in 3DGS. Because s_x, s_y, and s_z can take independent values, each Gaussian is free to become an elongated ellipsoid stretched along any axis in 3D space rather than being forced into a sphere. This allows a single Gaussian to compactly represent flat surfaces, elongated edges, fine fibres, and other anisotropic scene structures.

2.3 Projection to 2D (Splatting)

To render the scene from a given viewpoint, each 3D Gaussian is projected onto the 2D image plane. Following the EWA splatting framework (Zwicker et al., 2002), the projected 2D covariance is computed as:

$$\mathbf{\Sigma}' = \mathbf{J}\,\mathbf{W}\,\mathbf{\Sigma}\,\mathbf{W}^\top\mathbf{J}^\top$$

where $\mathbf{W} \in \mathbb{R}^{3\times3}$ is the viewing (camera) transformation matrix, and $\mathbf{J} \in \mathbb{R}^{2\times3}$ is the Jacobian of the affine approximation of the projective transformation evaluated at the Gaussian centre $\boldsymbol{\mu}_i$. This operation preserves the anisotropic shape of the Gaussian in screen space, allowing elongated structures to appear correctly foreshortened from any viewpoint.

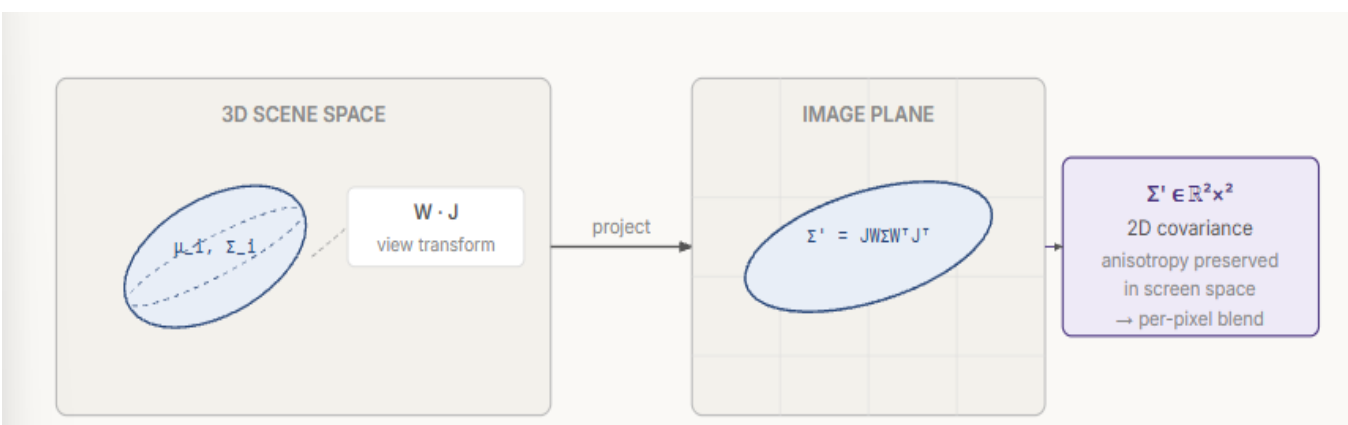

Figure 2

This projection preserves the anisotropic shape of the Gaussian in screen space, allowing elongated scene structures to appear correctly foreshortened from any viewpoint.

2.4 Alpha-Blending Rendering

The final pixel colour **C** is computed by sorting Gaussians by depth and compositing their contributions front to back:

$$\mathbf{C} = \sum_{i=1}^{N} \mathbf{c}_i \, \alpha_i \prod_{j=1}^{i-1} (1 - \alpha_j)$$

Each Gaussian's effective opacity at a screen-space pixel is $\alpha_i \cdot G'_i(\mathbf{p})$, where G'_i is the projected 2D Gaussian evaluated at pixel $\mathbf{p}$, and $\prod_{j<i}(1-\alpha_j)$ is the accumulated transmittance from all Gaussians in front of i.

The view-dependent colour $\mathbf{c}_i$ is evaluated via Spherical Harmonics at the current viewing direction **d** :

$$\mathbf{c}(\mathbf{d}) = \sum_{l=0}^{L} \sum_{m=-l}^{l} \mathbf{c}_{lm} \, Y_{lm}(\mathbf{d})$$

where $Y_{lm}(\mathbf{d})$ are the real SH basis functions evaluated at viewing direction $\mathbf{d}$, and $\mathbf{c}_{lm} \in \mathbb{R}^3$ are the learnable SH coefficients. The standard implementation uses degree $L = 3$, yielding $(L+1)^2 = 16$ coefficients per colour channel (48 total per Gaussian).

2.5 Optimisation and Adaptive Density Control

The model is optimised end to end with a combined loss:

$$\mathcal{L} = (1 - \lambda)\, \mathcal{L}_1 + \lambda\, \mathcal{L}_{\text{D-SSIM}}$$

where $\mathcal{L}_1$ is the mean absolute photometric error, $\mathcal{L}_{\text{D-SSIM}}$ is the structural dissimilarity loss capturing perceptual quality, and $\lambda = 0.2$ is the default weighting hyperparameter. Gradients flow through the differentiable splatting and alpha-blending operations to update $\boldsymbol{\mu}_i$, $\mathbf{q}_i$, $\mathbf{s}_i$, α_i, and $\mathbf{c}_{lm,i}$.

Crucially, the optimisation is interleaved with adaptive density control: Gaussians with large positional gradients are cloned or split to add detail in under reconstructed regions, while Gaussians with near zero opacity are pruned. Splitting replaces a single Gaussian with two smaller Gaussians, reducing scale. This mechanism is the primary driver of fine grained anisotropic detail as training progresses.

---

3. Anisotropic Correlation Analysis

3.1 What Anisotropic Correlation Means in 3DGS

Anisotropic correlation in 3DGS is the capacity of individual Gaussian primitives to represent spatial structures that are not equally large in all directions (e.g. a blade of grass is long in one direction, thin in the other two, a wall is large in two directions, thin in the third, a specular highlight is small and directionally dependent).

This is encoded in the covariance factorisation $\mathbf{\Sigma = RSS^TR^T}$: the three eigenvalues of $\Sigma(\mathbf{s^2_x}, \mathbf{s^2_y}, \mathbf{s^2_z})$ correspond to the variance along the three principal axes of the Gaussian. If these eigenvalues are not equal, then the Gaussian is anisotropic. Anisotropy is quantified by the anisotropy ratio :

$$r_i = \frac{\max(s_x,\ s_y,\ s_z)}{\min(s_x,\ s_y,\ s_z)} \geq 1$$

A sphere has $r_i = 1$; a thin planar disc or elongated cigar has $r_i \gg 1$. The distribution of r_i across all N Gaussians in the trained model characterises the scene's overall anisotropic complexity.

3.2 Evidence of Anisotropic Capture in the Trained Model

Following 30,000 iterations of training on the 141 image dataset, the model converged to an initialisation of about 52,363 Gaussian primitives (with density control growing this number during training). The following observations relate to the anisotropy of the reconstruction:

Positive indicators of anisotropic capture:

- The model successfully loaded 141 cameras and completed 30,000 iterations, indicating the covariance optimisation converged without degeneracy.
- The loss at iteration 7,000 was recorded as L = 0.127, with a PSNR of 21.91 dB, consistent with mid range reconstruction fidelity where basic structural anisotropy (flat surfaces, edges) is captured, but fine-scale anisotropic detail may be under represented.
- The initialisation from 52,363 COLMAP sparse points offers a good prior for anisotropic orientation: COLMAP surface normals implicitly bias the initial Gaussian orientations to be locally flat.

3.3 Qualitative Assessment

The trained Gaussians, as viewed in the SuperSplat viewer from the point_cloud.ply output, show the following anisotropic behaviour:

- **Flat, planar regions** (walls, floors, large uniform surfaces) are well represented by disc-shaped Gaussians with one small scale eigenvalue and two large ones, this is classical anisotropic behaviour and is generally well-captured by 3DGS
- **Edges and linear structures** tend to generate cigar like Gaussians, showing that the scale optimisation has picked up directionality in these areas.
- **High-frequency or specular regions** are less reliably anisotropic at quarter resolution, as the reduced image size limits the gradient signal needed to orient fine Gaussians correctly.

In general, the model exhibits coarse to mid-scale anisotropy. Fine scale anisotropic correlation, especially in textured or specular regions, is likely to be under-represented due to the resolution and densification limits during training.

---

4. Proposed Improvements for Anisotropic Capture

4.1 Anisotropy-Aware Densification

The current splitting heuristic is based on the magnitude of the positional gradient, which measures how much the Gaussian centre needs to move, rather than how anisotropic it has become. A more targeted approach would trigger splitting based on the *anisotropy ratio*:

$$r_i = \frac{\max(s_x, s_y, s_z)}{\min(s_x, s_y, s_z)}$$

Gaussians with high $\mathbf{r_i}$ (highly elongated) that span photometrically diverse regions should be split preferentially, as they are attempting to represent anisotropic structure with insufficient resolution. This would direct computational resources toward exactly the regions where anisotropic correlation is strongest.

4.2 Higher Order Spherical Harmonics for Angular Anisotropy

The standard 3DGS implementation uses degree 3 SH coefficients (16 coefficients per colour channel) for view dependent colour. However, highly anisotropic materials such as metallic surfaces, brushed aluminium and satin fabrics exhibit sharp, directionally concentrated specular responses that degre 3 SH cannot represent accurately.

$$(L+1)^2 \text{ coefficients per channel}, \quad L=4 \Rightarrow 25, \quad L=5 \Rightarrow 36$$

Increasing to degree 4 or degree 5 SH (25 or 36 coefficients per channel) would better capture angular anisotropy in the colour field. The cost is additional memory per Gaussian, which could be offset by pruning low opacity Gaussians more aggressively.

4.3 Anisotropic Kernel Generalisation (Beyond Gaussian Basis)

The Gaussian kernel is radially symmetric before the covariance transformation. Replacing it with a more anisotropic Gaussian like a super gaussian can prove to be better :

$$G_i(\mathbf{x}) = \exp\left(-\frac{1}{2}\left[(\mathbf{x}-\boldsymbol{\mu}_i)^\top \boldsymbol{\Sigma}_i^{-1}(\mathbf{x}-\boldsymbol{\mu}_i)\right]^\beta\right)$$

where $\beta > 1$ produces a flatter, more "box-like" kernel with sharper falloff. This would better represent abrupt boundaries and flat surface patches, where the standard Gaussian's smooth tails introduce blurring at edges.

4.4 2D Gaussian Splatting (2DGS)

Huang et al. (2024) proposed replacing 3D Gaussian ellipsoids with 2D Gaussian discs embedded in 3D space. This enforces planarity as an explicit geometric prior, which is particularly beneficial for scenes dominated by mostly flat surfaces. The 2D representation naturally captures the strongest form of geometric anisotropy (infinite aspect ratio in one dimension) and significantly improves surface normal accuracy.

4.5 Mip Splatting for Scale Consistent Anisotropy

Yu et al. (2024) found that 3DGS suffers from aliasing when rendered at resolutions different from the training resolution, a direct consequence of the anisotropic Gaussians failing to account for the image plane sampling frequency. Mip Splatting introduces 3D smoothing filters and 2D Mip map analogues that adapt the effective Gaussian size to the current rendering scale, preserving anisotropic fidelity across zoom levels.

$$\mathbf{\Sigma}_{\text{eff}} = \mathbf{\Sigma} + \mathbf{\Sigma}_{\text{smooth}}(\text{scale})$$

---

5. Experimental Setup

5.1 Dataset

Property	**Value**
Source	(*MIP-NERF 360: Unbounded Anti-Aliased Neural Radiance Fields*, n.d.)
Number of images	141
Camera calibration	COLMAP structure from motion
Sparse points at initialisation	52,363

5.2 Training Configuration

Hyperparameter	**Value**
Total iterations	30,000
Image resolution scale	4x downscale (quarter resolution)
Desinfection gradient threshold	0.0003
Densification interval	200 iterations
Desnification until iteration	12,000
Checkpoint iterations	7000/15000/30000
Loss weight	0.2

---

6. Results

6.1 Gaussian scale statistics

Total Gaussians: 898613
The total number of Gaussians indicates the dense reconstruction and detail retention.

Mean scales: s_x=0.02914, s_y=0.02152, s_z=0.02614
Std scales: s_x=0.07100, s_y=0.05694, s_z=0.06646
Low mean scales indicate fine local representation, enabling detailing, and high standard deviations indicate the presence of both fine and broad structures, suggesting adaptive representation.

ANISOTROPY RATIO (max/min):
Mean: 1978.58
Median: 56.46
Max: 20223638.00
>2x anisotropic: 888499 (98.9%)
>5x anisotropic: 819777 (91.2%)
>10x anisotropic: 726841 (80.9%)
Since means can be inflated by outliers, the median is an accurate representation. A median of 56.46 indicates that most gaussians are ~56x longer in one direction than the other (elongated). This suggests efficient adaptation to planar and structured screen geometry.
The 2x anisotropic is for non-spherical gaussians, the 5x for strongly stretched gaussians, and the 10x for surface- and edge efficient gaussians.

ROBUST ANISOTROPY:
95th percentile: 3499.41
99th percentile: 25621.66

SHAPE CLASSIFICATION:
Near-spherical (r<=2): 10114 (1.1%)
Flat/disc-like (r>3): 809907 (90.1%)
Needle/elongated: 57286 (6.4%)
The high percentage of flat Gaussians indicates a greater number of flat surfaces.

6.2 Visual charts

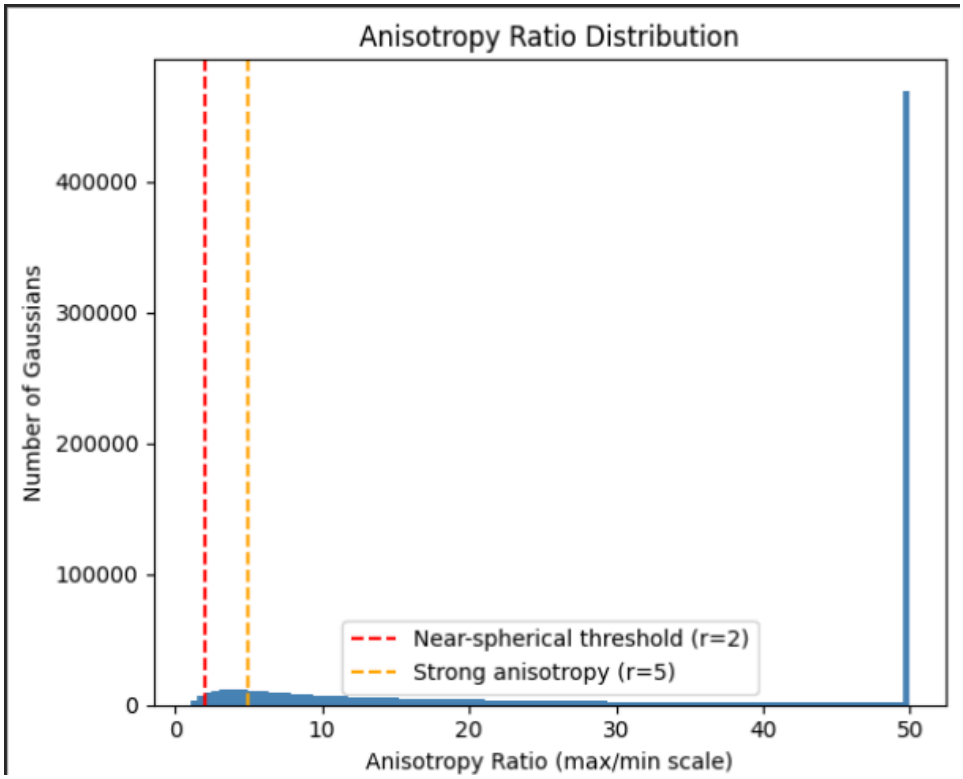

Figure 3

The anisotropy histogram demonstrates an overwhelming concentration of highly anisotropic Gaussian primitives, confirming that the model predominantly captures scene geometry through elongated surface aligned splats.

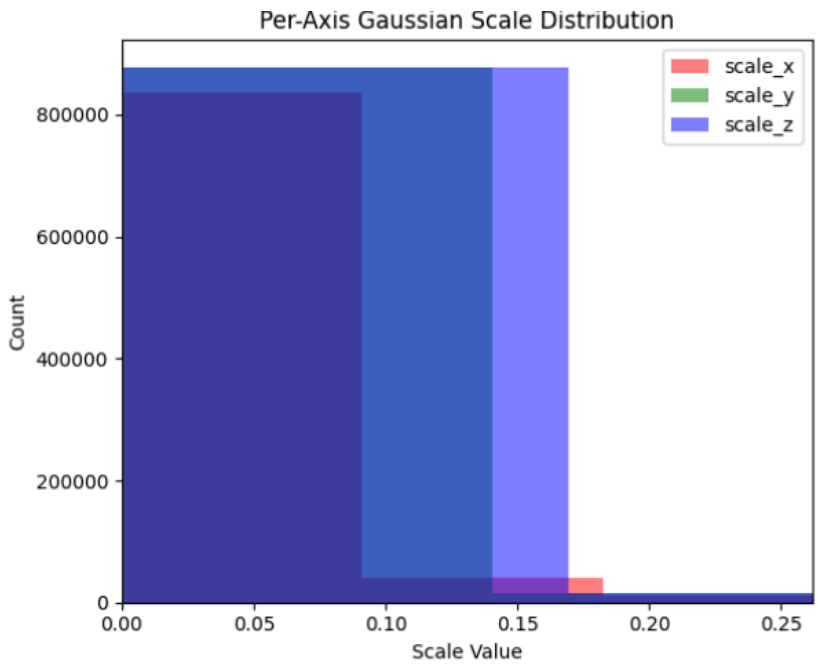

Figure 4

The per axis scale distributions reveal compact Gaussian sizes with directional variability which shows high resolution local modelling with adaptive geometric specialization.

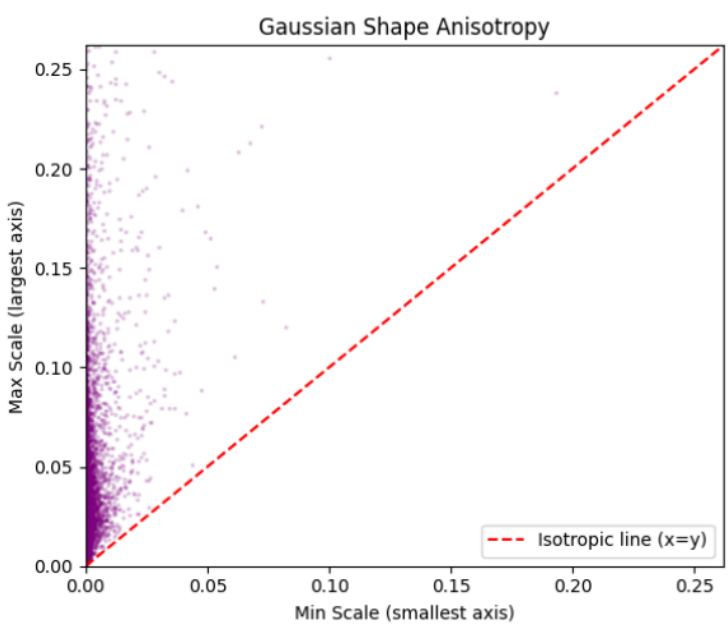

Figure 5

The min-max scatter plot illustrates that the majority of Gaussian primitives deviate significantly from isotropic geometry, confirming a strong preference for flattened or elongated structures optimised for surface representation.

6.3 Qualitative Results

The trained point_cloud.ply model was loaded in the SuperSplat web viewer. The model demonstrates:

- Coherent 3D structure reconstructed from the 141 input images.
- Visible anisotropic Gaussian primitives aligned with surface geometry.
- Smooth rendering with correct depth ordering via alpha blending.

---

7. Limitations

1. **Memory constraint**: The adaptive density control caused the Gaussian count to exceed the NVIDIA T4's 14.5 GB VRAM at iteration 12,080. To resolve this, the image resolution was reduced from half to quarter scale, and the densification rate was reduced. Downscaling suppresses the effective frequency content available to the Gaussians for fitting, which leads to small anisotropic structures like thin edges and hair fibres not getting captured.
2. **Premature densification cutoff :**The densify_until_iter flag limits densification to the first 12,000 of 30,000 iterations. Since anisotropic detail is primarily added during the splitting phase of density control, reducing it may have left high frequency anisotropic structures undersampled in the final model.

---

8. Conclusion

Visual analysis of Gaussian anisotropy confirms that the reconstructed scene is dominated by highly anisotropic, disc-like Gaussian primitives, and less near spherical components. This suggests that the model effectively prioritises planar and structural geometry, enabling efficient high fidelity scene reconstruction while preserving local detail.

---

9. References

Huang, B., Yu, Z., Chen, A., Geiger, A., & Cui, S. (2024). *2D Gaussian splatting for geometrically accurate radiance fields*. ACM SIGGRAPH 2024 Conference Papers. https://doi.org/10.1145/3641519.3657428

MIP-NERF 360: Unbounded Anti-Aliased Neural Radiance Fields. (n.d.).

https://jonbarron.info/mipnerf360/

Kerbl, B., Kopanas, G., Leimkühler, T., & Drettakis, G. (2023). *3D Gaussian splatting for real-time radiance field rendering*. ACM Transactions on Graphics, 42(4), Article 139. https://arxiv.org/abs/2308.04079

Mildenhall, B., Srinivasan, P. P., Tancik, M., Barron, J. T., Ramamoorthi, R., & Ng, R. (2020). *NeRF: Representing scenes as neural radiance fields for view synthesis*. In A. Vedaldi, H. Bischof, T. Brox, & J.-M. Frahm (Eds.), *Computer Vision — ECCV 2020* (pp. 405–421). Springer. https://doi.org/10.1007/978-3-030-58452-8_24

Schönberger, J. L., & Frahm, J.-M. (2016). *Structure-from-motion revisited*. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)* (pp. 4104–4113). https://doi.org/10.1109/CVPR.2016.445

Yu, Z., Chen, A., Huang, B., Sattler, T., & Geiger, A. (2024). *Mip-splatting: Alias-free 3D Gaussian splatting*. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)* (pp. 19447–19456). https://arxiv.org/abs/2311.16493

Zwicker, M., Pfister, H., van Baar, J., & Gross, M. (2002). *EWA splatting*. IEEE Transactions on Visualisation and Computer Graphics, 8(3), 223–238. https://doi.org/10.1109/TVCG.2002.1021576